

Mike Y. M. Lau

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Heidelberg Institute for Theoretical Studies (HITS)
Centre for Astronomy (ZAH), Heidelberg University

Education

- 09/19 – 03/23 **Monash University**, Melbourne, Australia (PhD in Astrophysics, conferred Aug 2023)
Dissertation title: *Interactions in Stellar Binaries*
Supervisors: Ilya Mandel, Daniel J. Price, & Ryosuke Hirai
- 10/15 – 07/19 **University of Oxford**, Oxford, United Kingdom (MMathPhys, First Class & Distinction)
Master of Mathematical and Theoretical Physics with First Class in Parts A, B, & C
Dissertation title: *Detecting Double Neutron Stars with LISA*
Supervisors: Ilya Mandel & Philipp Podsiadlowski

Academic experience

- 09/23 – **Heidelberg Institute for Theoretical Studies**, Heidelberg, Germany
09/23–08/25: Croucher Research Fellow
09/25–Present: Postdoctoral researcher at Astronomisches Rechen-Institut (ARI), Centre for Astronomy of Heidelberg University (group of Friedrich K. Röpke)
- 01 – 06/22 **Flatiron Institute Center for Computational Astrophysics**, New York, United States
(Research Analyst in pre-doctoral program)
Supervisors: Matteo Cantiello & Adam Jermyn

Grants & awards

- | | | |
|---------|---|--|
| 24 | Shortlist for Mollie Holman Doctoral Medal for Science | <i>Monash University</i> |
| 04/23 | Croucher Research Fellowship | <i>Croucher Foundation</i> |
| 03/23 | Humboldt Research Fellowship (declined for Croucher) | <i>Alexander von Humboldt Foundation</i> |
| 01/23 | Postgraduate publication award | <i>Monash University</i> |
| 12/22 | Max Planck Institute for Astrophysics Fellowship (declined for Croucher) | <i>Max Planck Institute for Astrophysics</i> |
| 21 | Lead CI for NCI Astronomy Program computing grant
Q1/2: 544 kSU, Q3/4: 670 kSU | <i>AAL Astronomy Supercomputer</i> |
| 19 – 22 | J. L. William International PhD Scholarship | <i>Monash University</i> |
| 19 – 23 | Research Training Program (RTP) Stipend | <i>Monash University</i> |
| 19 – 23 | Monash International Tuition Scholarship | <i>Monash University</i> |
| 07/19 | Schools Prize | <i>St Edmund Hall, University of Oxford</i> |
| 17, 18 | Open Scholarship | <i>St Edmund Hall, University of Oxford</i> |
| 16 | Open Exhibition | <i>St Edmund Hall, University of Oxford</i> |
| 08/15 | Hong Kong Scholarship for Excellence | <i>Hong Kong Government</i> |

Teaching & supervision

- 26 – Pranavandhan Upendranath (Heidelberg University), Masters thesis project co-supervision with Fabian Schneider, simulations of stellar mergers
- 24 – 26 Ana Lourdes (Macquarie University), PhD co-supervision with Orsola De Marco, Thesis title: *Accretion disk simulations in pre-common envelope binary systems*
- 23 – Project co-supervision for PhD & Master's students at the Heidelberg Institute for Theoretical Studies
- 09/26 Invited lecturer, 21st Heidelberg Summer School, *Cosmic Multiplicity: Evolution and Fate of Binary Stars*
- 11 – 12/21 Co-supervisor for summer Honours student at Monash University
- TA *ASP3051 Relativity and Cosmology* (02–06/21), *ASP1010: Earth to Cosmos—Introductory Astronomy* (02–06/20)
- Tutor for indigenous access programme *ASP3162 Computational Astrophysics and the Extreme Universe* (08–11/20), *ASP3012 Stars and Galaxies* (08–11/20), *MCD1180: Introductory Physics* (08–09/20), *ASP3051 Relativity and Cosmology* (04–06/20), *MAT9004 Mathematical Foundations for Data Science* (04–06/20)

Academic service & outreach

Journal review	<i>The Astrophysical Journal</i> (Letters and Main Journal), <i>Monthly Notices of the Royal Astronomical Society</i> , <i>Astronomy and Astrophysics</i> , <i>Physical Review D</i> , & <i>The Open Journal of Astrophysics</i>
Proposal review	JWST Director's Discretionary Time, National Science Centre Poland
02/23	LOC, 2023 Phantom Users Workshop
11/22	OzGrav Outreach Superstar Award
10/19 – 08/23	Organiser, weekly Whiteboard Seminars at Monash University
09/22	World Science Festival, Ipswich, Queensland
07/21	Dark Science holiday programme, Casey Tech School (Berwick)
07/21	Black Hole Sunday, TwistED Science Centre, Echuca, Victoria
04/21	OzGrav Interactive tech showcase, Bendigo Discovery Science and Technology Centre
12/19	Monash Minimaker Faire, Monash University
18	Organiser for St Edmund Hall Physics Journal Club
08/17 – 19	Academic and Scholarship Mentor, Project Access HK

Publications

Submitted works (1 second-author)

26. Juarez-Garcia, A., **Lau, M.**, De Marco, O., Siess, L., Price, D. et al., 2026, *Three-dimensional simulations of accretion disks in pre-CE systems – II. Accretion efficiency and angular momentum transport*, submitted, [arXiv:2609.17998](#)
25. Mu, C., De Marco, O., Bermúdez-Bustamante, L., Hirai, R., Price, D. et al. (inc. **Lau, M.**), 2026, *Dust Formation in Common Envelope Binary Interactions—III. Lightcurves*, submitted, [arXiv:2605.26495](#)
24. Bernard, Y., David-Cléris, T., Price, D., **Lau, M.**, 2026, *Momentum-conserving self-gravity in the phantom smoothed particle hydrodynamics code. Parallel dual tree traversal for the symmetric fast multipole method*, submitted to A&A, [arXiv:2602.05804](#)
23. Heller, M., Schneider, F., Henneco, J., Bronner, V., **Lau, M.**, 2025, *1D stellar mergers: entropy sorting and PyMMAMS*, submitted, [arXiv:2512.14669](#)

Published/accepted works (7 first-author, 2 second-author)

22. **Lau, M.**, Andrassy, R., Leidi, G., Gagnier, D., Morán-Fraile, J. et al., 2026, *Continuous mass ablation of planets engulfed in stellar envelopes*, accepted by A&A, [arXiv:2605.11343](#)
21. Wang, C., **Lau, M.**, Li, X., Langer, N., de Mink, S. et al., 2026, *Thermal-timescale accretion does not always yield critical rotation in mass gainers*, *A&A*, 707
20. Vetter, M., Röpke, F., Schneider, F., Pakmor, R., Ohlmann, S. et al. (inc. **Lau, M.**), 2025, *Magnetically driven outflows in the 3D common envelope evolution of massive stars*, *A&A*, 698
19. Bermúdez-Bustamante, L., De Marco, O., Siess, L., Price, D., González-Bolívar, M. et al. (inc. **Lau, M.**), 2025, *Dust formation during the interaction of binary stars by common envelope*, *Planetary Nebulae (IAU S384) A Universal Toolbox in the Era of Precision Astrophysics, Proceedings of the IAU Symposia and Colloquia*
18. **Lau, M.**, Hirai, R., Price, D., Mandel, I., Bate, M., 2025, *Common envelopes in massive stars: III. The obstructive role of radiation transport in envelope ejection*, *A&A*, 699
17. Mandel, I., Riley, J., Boesky, A., Brcek, A., Hirai, R. et al. (inc. **Lau, M.**), 2025, *Rapid stellar and binary population synthesis with COMPAS: methods paper II*, *ApJS*, 280, 1
16. Vetter, M., Röpke, F., Schneider, F., Pakmor, R., Ohlmann, S. et al. (inc. **Lau, M.**), 2025, *Magnetically driven outflows in the 3D common envelope evolution of massive stars*, *A&A*, 698
15. Siess, L., Bermúdez-Bustamante, L., De Marco, O., Price, D., González-Bolívar, M. et al. (inc. **Lau, M.**), 2024, *Dusty Common Envelope Evolution*, *Galaxies*, 12
14. Schneider, F., **Lau, M.**, Röpke, F., 2025, *Stellar mergers and common-envelope evolution*, *Encyclopedia of Astrophysics - 1st Edition*, Elsevier
13. Vetter, M., Röpke, F., Schneider, F., Pakmor, R., Ohlmann, S. et al. (inc. **Lau, M.**), 2024, *From spherical stars to disk-like structures: 3D common-envelope evolution of massive binaries beyond inspiral*, *A&A*, 691
12. Bermúdez-Bustamante, L., De Marco, O., Siess, L., Price, D., González-Bolívar, M. et al. (inc. **Lau, M.**), 2024, *Dust formation in common envelope binary interactions - II: 3D simulations with self-consistent dust formation*, *MNRAS*, 533, 1
11. **Lau, M.**, Hirai, R., Mandel, I., Tout, C., 2024, *Expansion of Accreting Main-sequence Stars during Rapid Mass Transfer*, *ApJL*, 966, 1
10. Amaro-Seoane, P., Andrews, J., Arca Sedda, M., Askar, A., Baghi, Q. et al. (inc. **Lau, M.**), 2023, *Astrophysics with the Laser Interferometer Space Antenna*, *Living Reviews in Relativity*, 26, 1

9. Renzo, M., Zapartas, E., Justham, S., Breivik, K., **Lau, M.** et al., 2023, *Rejuvenated Accretors Have Less Bound Envelopes: Impact of Roche Lobe Overflow on Subsequent Common Envelope Events*, *ApJL*, 942, 2
8. González-Bolívar, M., De Marco, O., **Lau, M.**, Hirai, R., Price, D., 2022, *Common envelope binary interaction simulations between a thermally pulsating AGB star and a low mass companion*, *MNRAS*, 517, 3
7. **Lau, M.**, Hirai, R., Price, D., Mandel, I., 2022, *Common envelopes in massive stars II: The distinct roles of hydrogen and helium recombination*, *MNRAS*, 516, 4
6. **Lau, M.**, Cantiello, M., Jermyn, A., MacLeod, M., Mandel, I. et al., 2025, *Hot Jupiter engulfment by an early red giant in 3D hydrodynamics*, *A&A*, 694
5. **Lau, M.**, Hirai, R., González-Bolívar, M., Price, D., De Marco, O. et al., 2022, *Common envelopes in massive stars: towards the role of radiation pressure and recombination energy in ejecting red supergiant envelopes*, *MNRAS*, 512, 4
4. Riley, J., Agrawal, P., Barrett, J., Boyett, K., Broekgaarden, F. et al. (inc. **Lau, M.**), 2022, *Rapid Stellar and Binary Population Synthesis with COMPAS*, *ApJS*, 258, 2
3. Compas, T., Riley, J., Agrawal, P., Barrett, J., Boyett, K. et al. (inc. **Lau, M.**), 2022, *COMPAS: A rapid binary population synthesis suite*, *The Journal of Open Source Software*, 7, 69
2. Ackley, K., Adya, V., Agrawal, P., Altin, P., Ashton, G. et al. (inc. **Lau, M.**), 2020, *Neutron Star Extreme Matter Observatory: A kilohertz-band gravitational-wave detector in the global network*, *Publications of the Astronomical Society of Australia*, 37
1. **Lau, M.**, Mandel, I., Vigna-Gómez, A., Neijssel, C., Stevenson, S. et al., 2020, *Detecting double neutron stars with LISA*, *MNRAS*, 492, 3

Selected talks (* = invited)

09/26	*21st Heidelberg Summer School (invited lecture)	Heidelberg
12/25	XIX Winter Workshop on Stellar Astrophysics 2025	HITS, Heidelberg
11/25	Hong Kong Laureate Forum	Hong Kong
09/25	*Astronomy seminar	Tsung-Dao Lee Institute
08/25	*Binary Stars in a New Era	Lijiang
06/25	*European Astronomical Society Annual Meeting (invited review)	Cork
06/25	*2nd European Phantom code family users workshop	IPAG, Grenoble
12/24	*Astrophysics seminar	Chinese University of Hong Kong
07/24	41st Liège International Astrophysical Colloquium: The eventful life of massive star multiples (best linguistics invention)	University of Liège
02/24	*Joint Franco-Australian 5th Phantom and MCFOST Users Workshop	Monash University, remote
12/23	*Astrophysics seminar	Chinese University of Hong Kong
03/23	*Colloquium	ICRAR-Curtin, Perth
02/23	Phantom users workshop 2023 (LOC)	Monash University
01/23	SESTAS meeting	MPA, Garching
01/23	Common envelope group meeting	HITS, Heidelberg
12/22	*Gravitational Wave Physics and Astronomy Workshop	Melbourne
06/22	CCA Predoctoral Program Symposium	CCA, Flatiron Institute
06/22	Physics and Astrophysics of Common Envelope	Los Alamos National Laboratory
03/22	CCA Stars & Compact Objects Group Meeting	CCA, Flatiron Institute
12/21	OzGrav Data/Astro Telecon	Virtual
09/21	Common Envelope Physics and Outcomes 2021	Virtual
07/21	ASA Annual Meeting 2021	University of Melbourne
07/21	EAS Annual Meeting 2021	Leiden, virtual
02/21	*LISA Workshop	University of Auckland, remote
08/20	The 13th International LISA Symposium	University of Auckland, remote
02/20	ANITA workshop and school 2020	UNSW, Canberra
01/20	Gravitational Waves Group Meeting	Cardiff University
01/20	Astrophysics Seminar	University of Birmingham
12/19	2019 Stars in Melbourne	Monash University
11/19	2019 OzGrav Annual Retreat	Lorne, Melbourne
11/19	OzGrav Data/Astro Telecon	Virtual

Miscellaneous

- Software development: COMPAS (rapid binary population synthesis), PHANTOM (smoothed particle hydrodynamics)
- User: MESA (stellar evolution), REBOUND (N-body integrator), Seven-League Hydro (finite-volume hydrodynamics), Phlegethon (finite-volume hydrodynamics), Athena++ (finite-volume hydrodynamics)
- Programming languages: Python, Fortran, C++, MATLAB
- Languages: English (native), Cantonese (native), Mandarin (native), German (basic)